

Notes: Bolton, Chen, and Wang (JF, 2011)
A Unified Theory of Tobin's q , Corporate Investment, Financing, and Risk Management

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1 Big Picture

This note summarizes Bolton, Chen, and Wang (2011), which builds a continuous-time dynamic model of a financially constrained firm jointly choosing investment, cash holdings, external financing, payout, hedging, and credit-line usage. The key economic object is the endogenous marginal value of liquidity. Once external financing is costly, investment is no longer governed by marginal q alone; instead, investment depends on the ratio of marginal q to the marginal value of liquidity.

1.1 One-sentence summary

The paper provides a unified dynamic corporate finance theory in which financially constrained firms manage real investment, cash, equity issuance, payout, derivatives hedging, and credit lines through the marginal value of liquidity, showing that cash and hedging are complementary risk-management tools and that Tobin's q has different investment implications depending on the firm's marginal source of funding.

1.2 Core question

The paper asks: how should a firm make investment, financing, payout, cash-management, and risk-management decisions when external funds are costly and cash is valuable? More specifically, it asks how a firm should choose:

- investment when internal liquidity is scarce;
- cash reserves and payout boundaries;
- whether and when to issue equity;
- whether to hedge systematic risk using derivatives;
- how credit lines change the link between q and investment.

1.3 Main economic answer

The answer is organized around the marginal value of cash, $p'(w)$, where $w = W/K$ is the firm's cash-capital ratio. When $p'(w)$ is high, cash is very valuable and the firm cuts investment, sells assets, delays payout, hoards liquidity, and may hedge more aggressively if hedging is feasible. When $p'(w)$ is close to one, the firm behaves more like the frictionless neoclassical benchmark.

2 Incremental Contribution of the Paper

2.1 Incremental contribution relative to neoclassical q theory

Traditional q theory says investment should be linked to marginal q . In the frictionless Hayashi setting, average q and marginal q coincide, and investment depends on marginal q alone. Bolton, Chen, and Wang change this logic by adding costly external finance and liquidity management.

Their investment first-order condition becomes:

$$1 + \theta i(w) = \frac{q^m(w)}{p'(w)}, \quad (1)$$

where $q^m(w)$ is marginal q and $p'(w)$ is the marginal value of liquidity. Thus, even if marginal q is high, investment can be low when liquidity is scarce and $p'(w)$ is high.

2.2 Incremental contribution relative to the financing-constraints literature

Earlier empirical work, such as Fazzari, Hubbard, and Petersen (1988), uses investment-cash flow sensitivity to diagnose financing constraints, while Kaplan and Zingales (1997) criticizes the interpretation of such sensitivities. This paper gives a dynamic structural foundation for why cash, investment, and financing constraints interact nonlinearly. It shows that the firm may appear unconstrained most of the time because it optimally holds cash to avoid constrained states, even though rare bad states still strongly shape firm value and investment policy.

2.3 Incremental contribution relative to static risk-management theory

Froot, Scharfstein, and Stein (1993) emphasize hedging when external finance is costly. Bolton, Chen, and Wang extend this idea dynamically. Risk management is not only derivatives hedging: it also includes cash accumulation, asset sales, investment cuts, payout timing, and credit-line usage. The paper's central novelty is showing that liquidity management and financial hedging are complementary, not simple substitutes.

2.4 Incremental contribution relative to cash-holding models

Cash-holding models often focus on a target cash ratio. This paper shows that the optimal policy is better described as a double-barrier policy. The firm pays out cash at an endogenous upper boundary and raises external finance or liquidates at a lower boundary. Between the two boundaries, the firm continuously manages investment, asset sales, and hedging as a function of liquidity.

2.5 Why the paper matters

The paper matters because it links several corporate finance margins that are often studied separately. It says that investment, cash, external financing, payout, hedging, and credit-line policies are not independent decisions. They are all responses to the same state variable: financial slack or liquidity.

3 Model Setup

3.1 Production technology

The firm uses capital K_t to produce. Capital evolves as

$$dK_t = (I_t - \delta K_t)dt, \quad (2)$$

where I_t is gross investment and δ is depreciation. The firm's risk-adjusted productivity shock follows

$$dA_t = \mu dt + \sigma dZ_t. \quad (3)$$

Operating cash flow is

$$dY_t = K_t dA_t - I_t dt - G(I_t, K_t) dt. \quad (4)$$

The adjustment cost is homogeneous of degree one:

$$G(I, K) = g(i)K, \quad i = I/K, \quad g(i) = \frac{\theta i^2}{2}. \quad (5)$$

Read:

- The model uses an AK-style technology, so investment opportunities are constant over time.
- This is deliberate: the authors want to isolate financing frictions, not time-varying productivity or growth options.
- Homogeneity allows the two-state problem in (K, W) to collapse into a one-state problem in $w = W/K$.

Economic message: The model abstracts from changing investment opportunities so that all dynamics come from liquidity and financing constraints. This makes the marginal value of cash the key state-dependent object.

3.2 External financing costs

The firm can raise external equity but must pay both fixed and proportional financing costs:

- fixed cost: ϕK ;
- proportional cost: γ per dollar raised.

The fixed cost is scaled by capital to preserve homogeneity. This means equity issuance occurs infrequently and in lumps.

3.3 Cash accumulation

Let W_t be cash. The cash accumulation equation is

$$dW_t = dY_t + (r - \lambda)W_t dt + dH_t - dU_t, \quad (6)$$

where λ is the cash-carrying cost, H_t is cumulative external financing, and U_t is cumulative payout.

Read:

- Cash earns less than the risk-free rate because of agency costs or tax costs of holding cash.
- Paying out cash avoids this carry cost, but reduces financial slack.
- Holding cash protects the firm from costly equity issuance, underinvestment, asset sales, or liquidation.

Economic message: Cash has option value. A dollar inside the firm is more valuable when the firm is close to financial distress, because it helps avoid costly external financing or liquidation.

4 Neoclassical Benchmark

Without financing frictions, the firm has a constant first-best investment-capital ratio:

$$i^{FB} = r + \delta - \sqrt{(r + \delta)^2 - 2(\mu - (r + \delta))/\theta}. \quad (7)$$

Tobin's q is

$$q^{FB} = 1 + \theta i^{FB}. \quad (8)$$

Read:

- In the frictionless benchmark, average q equals marginal q .
- Cash holdings and idiosyncratic risk do not affect investment.
- The firm invests only according to productivity, adjustment costs, and discounting.

Economic message: The neoclassical benchmark gives the clean comparison point: once financing costs are introduced, deviations from i^{FB} and q^{FB} measure the real effects of financial constraints.

5 Dynamic Firm Problem and Core Mechanism

5.1 Firm value and homogeneity

Firm value is written as

$$P(K, W) = Kp(w), \quad w = W/K. \quad (9)$$

This transforms the problem from two state variables to one state variable. The marginal value of cash is

$$P_W(K, W) = p'(w), \quad (10)$$

and marginal q is

$$P_K(K, W) = p(w) - wp'(w). \quad (11)$$

5.2 Investment condition

The investment-capital ratio satisfies

$$i(w) = \frac{1}{\theta} \left(\frac{p(w)}{p'(w)} - w - 1 \right). \quad (12)$$

Equivalently,

$$1 + \theta i(w) = \frac{q^m(w)}{p'(w)}. \quad (13)$$

Read:

- $q^m(w)$ captures the marginal value of installed capital.
- $p'(w)$ captures the marginal cost of financing investment using scarce liquidity.
- Investment falls when $p'(w)$ rises, even if investment opportunities are unchanged.

Economic message: Financing constraints operate like an endogenous shadow cost of funds. The firm invests less not because projects are worse, but because internal liquidity is more valuable.

5.3 Double-barrier policy

The firm has three regions:

1. **External funding/liquidation region:** when w is too low, the firm raises equity or liquidates.
2. **Internal financing region:** when cash is between the lower and upper boundaries, the firm uses cash and adjusts investment.
3. **Payout region:** when w reaches the upper boundary $\bar{w}$, the firm pays out excess cash.

At the payout boundary,

$$p'(\bar{w}) = 1, \quad p''(\bar{w}) = 0. \quad (14)$$

At equity issuance, if the firm refinances rather than liquidates,

$$p(0) = p(m) - \phi - (1 + \gamma)m, \quad (15)$$

with

$$p'(m) = 1 + \gamma. \quad (16)$$

Economic message: The firm does not have a single target cash level. Instead, it has a liquidity band: retain cash inside the band, pay out at the upper boundary, and raise funds or liquidate at the lower boundary.

6 Main Results

6.1 Result 1: Investment depends on marginal q divided by the marginal value of liquidity

The first key result is the modified q theory:

$$\text{investment intensity } i(w) \text{ depends on } \frac{q^m(w)}{p'(w)}. \quad (17)$$

When the firm is flush with cash, $p'(w) \approx 1$, so the firm behaves close to the neoclassical benchmark. When the firm is liquidity constrained, $p'(w) > 1$, so investment is depressed.

Key takeaway: The same marginal q implies different investment levels depending on the firm's liquidity state.

6.2 Result 2: Equity issuance and payout follow an endogenous double-barrier policy

The firm waits to issue equity until it runs out of cash because issuing equity earlier has no benefit when investment is smooth and cash carries a cost. If it issues equity, it raises funds in a lump to economize on the fixed issuance cost. If cash is abundant, the firm pays out cash at the upper boundary.

Key takeaway: The model generates a dynamic pecking order: internal cash first, then credit line if available, then external equity.

6.3 Result 3: Cash management and derivatives hedging are complementary

The firm can reduce systematic risk through financial hedging and idiosyncratic risk through cash and investment policy. When derivatives hedging is costly because of margin requirements, the firm hedges less and relies more on cash.

Key takeaway: Hedging does not replace cash. Hedging and cash jointly manage risk, especially when external financing costs are important.

7 Quantitative Analysis

7.1 Benchmark parameters

The benchmark calibration uses annualized values: risk-free rate $r = 6\%$, depreciation $\delta = 10.07\%$, risk-neutral mean productivity $\mu = 18\%$, productivity volatility $\sigma = 9\%$, adjustment cost parameter $\theta = 1.5$, cash-carrying cost $\lambda = 1\%$, proportional financing cost $\gamma = 6\%$, fixed financing cost $\phi = 1\%$, and liquidation value $l = 0.9$.

Read:

- The first-best investment-capital ratio is $i^{FB} = 15.1\%$.
- First-best Tobin's q is $q^{FB} = 1.23$.
- These benchmarks are used to measure how much financial constraints depress investment and firm value.

7.2 Case I: liquidation

If the firm must liquidate when cash reaches zero, the marginal value of cash becomes extremely high near the lower boundary. In the calibration, an extra dollar of cash can be worth as much as about 30 dollars to the firm near zero cash.

Read:

- Firm value is concave in cash.
- The marginal value of cash explodes near the liquidation boundary.
- Investment falls far below first-best and can become negative, meaning asset sales.

Economic message: Severe financing constraints make firms aggressively preserve liquidity. The firm cuts investment and sells assets not because projects are unproductive, but because survival liquidity is extremely valuable.

7.3 Case II: refinancing

When external equity issuance is available, the firm prefers refinancing over liquidation under the benchmark parameters. The payout boundary falls relative to the liquidation case because future financing is possible. But fixed issuance costs still make cash valuable near the lower boundary.

Read:

- With fixed issuance cost $\phi = 1\%$, the firm raises cash in lumps.
- The marginal value of cash at zero cash is around 1.7, above the proportional issuance cost $1 + \gamma = 1.06$.
- Removing fixed financing costs greatly flattens the marginal value of cash and investment policy.

Economic message: Even moderate fixed financing costs can create strong precautionary cash demand and substantial underinvestment when cash is low.

8 Average q , Marginal q , and Investment

The firm's enterprise value is $P(K, W) - W$, so average q is

$$q^a(w) = \frac{P(K, W) - W}{K} = p(w) - w. \quad (18)$$

Marginal q is

$$q^m(w) = p(w) - wp'(w) = q^a(w) - (p'(w) - 1)w. \quad (19)$$

Read:

- With financing constraints, marginal q is generally below average q in the cash region.
- The gap reflects the fact that more capital mechanically lowers the cash-capital ratio.
- Thus, marginal capital has a financing-cost effect in addition to a production effect.

Economic message: Average and marginal q are no longer interchangeable. The standard Hayashi equivalence breaks down once liquidity affects firm value.

9 Stationary Distribution and Interpretation

The paper simulates the stationary distribution of cash holdings and policy variables. The firm spends most of its time with relatively high cash holdings and marginal value of cash close to one. This might make the firm look unconstrained in ordinary data. However, low-probability bad states still have strong effects on average firm value and investment.

Read:

- Mean cash-capital ratio is about 0.159 and median is about 0.169.
- Mean investment-capital ratio is about 0.104, much lower than the first-best 0.151.
- The marginal value of cash has a mean near one but extreme skewness and kurtosis.

Economic message: Financing constraints are hidden in the tails. Firms manage cash to avoid bad states, so observed average policies may understate the true importance of financing frictions.

10 Dynamic Hedging

10.1 Why hedging matters

The firm can hedge systematic risk using futures. If hedging has no margin friction, the firm can eliminate its systematic risk exposure. With margin requirements, hedging capacity depends on cash because the firm must post collateral.

10.2 Hedging regions

The model generates three hedging regions:

1. **Low-cash region:** margin constraints bind, so the firm hedges as much as cash allows.
2. **Middle region:** the firm chooses an interior hedge ratio.
3. **High-cash region:** hedging benefits are small, so the firm does not hedge.

Read:

- Hedging is most valuable when cash is scarce, but that is exactly when margin constraints bind.
- Hedging can increase firm value and allow higher investment when the firm is not severely constrained.
- In very low-cash states, hedging can raise the marginal value of cash and induce stronger precautionary saving.

Economic message: Risk management is state dependent. The firm may want to hedge most when it can least afford collateral. This links derivatives hedging directly to liquidity management.

11 Credit Lines

Credit lines provide another source of liquidity. The firm can draw up to cK , where c is a credit-line limit as a fraction of capital. The firm pays a spread a over the risk-free rate on drawn credit.

11.1 Pecking order with credit lines

The model generates the following financing hierarchy:

$$\text{cash} \rightarrow \text{credit line} \rightarrow \text{external equity}. \quad (20)$$

The firm uses internal cash first because drawn credit is costly. It uses the credit line before equity because equity issuance has fixed and proportional costs.

11.2 Credit lines and investment

Credit lines reduce the marginal value of cash and mitigate underinvestment. In the benchmark, access to a 20% credit line reduces the payout boundary and greatly softens investment cuts near zero cash.

Read:

- With a credit line, the payout boundary falls from 0.19 to 0.08.
- The marginal value of cash at $w = 0$ falls from about 1.69 without a credit line to about 1.01 with a credit line.
- Investment at zero cash improves from severe asset sales to positive investment.

Economic message: Credit lines are valuable because they substitute for costly equity issuance and reduce the need to hoard cash. However, they also change the meaning of marginal q .

12 Important Subtle Result: Marginal q Can Move Opposite to Investment

With a credit line, marginal q can decrease while investment increases. In the credit region, increasing capital relaxes leverage by increasing collateral capacity. Therefore, marginal capital has an additional debt-capacity benefit. This can make marginal q high when leverage is high, even though the firm invests little.

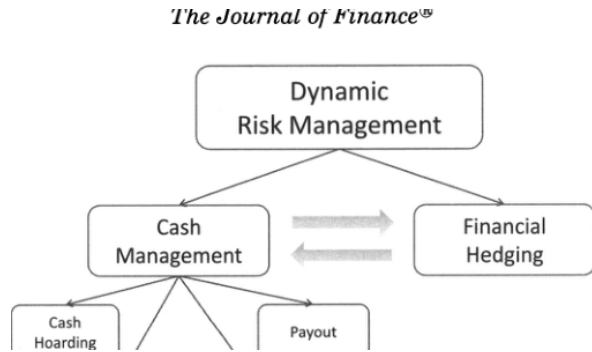
Read:

- In the cash region, investment and marginal q tend to move together.
- In the credit region, investment and marginal q can move in opposite directions.
- Average q may be a more robust empirical predictor of investment in the presence of credit lines.

Economic message: The paper warns against mechanically interpreting marginal q as an investment signal when financing constraints and credit capacity matter.

13 Extracted Figures and Tables from the Paper

13.1 Figure 1: A Unified Framework for Risk Management and Table I: Summary of Key Variables and Parameters



(a) Figure 1: A unified framework for risk management

This table summarizes the symbols for the key variables used in the model and the parameter values in the benchmark case. For each upper-case variable in the left column (except K , A , and F), we use its lower case to denote the ratio of this variable to capital. All the boundary variables are in terms of the cash-capital ratio w_t .

Variable	Symbol	Parameters	Symbol	Value
<i>Baseline Model</i>				
Capital stock	K	Risk-free rate	r	6%
Cash holding	W	Rate of depreciation	δ	10.07%
Investment	I	Risk-neutral mean productivity shock	μ	18%
Cumulative productivity shock	A	Volatility of productivity shock	σ	9%
Investment adjustment cost	G	Adjustment cost parameter	θ	1.5
Cumulative operating profit	Y	Proportional cash-carrying cost	λ	1%
Cumulative external financing	H	Capital liquidation value	l	0.9
Cumulative external financing cost	X	Proportional financing cost	γ	6%
Cumulative payout	U	Fixed financing cost	ϕ	1%
Firm value	P	Market Sharpe ratio	η	0.3
Average q	q_a	Correlation between market and firm	ρ	0.8
Marginal q	q_m			
Payout boundary	$\bar{w}$			
Financing boundary	$\underline{w}$			
Return cash-capital ratio	m			
<i>Hedging</i>				
Hedge ratio	ψ	Margin requirement	π	5
Fraction of cash in margin account	κ	Flow cost in margin account	ϵ	0.5%
Futures price	F	Market volatility	σ_m	20%
Maximum-hedging boundary	w_-			
Zero-hedging boundary	w_+			
<i>Credit Line</i>				
		Credit line limit	c	20%
		Credit line spread over r	α	1.5%

condition (19), (iv) the optimality condition for equity issuance (20), and (v) the endogenous payout boundary conditions (16) and (17). Finally, to verify that refinancing is indeed the firm's global optimal solution, it is sufficient to check

(b) Table I: Summary of key variables and parameters

Read:

- Figure 1 organizes dynamic risk management into cash management and financial hedging.
- Cash management includes cash hoarding, real investment, payout, and external financing.
- Table I lists the key model variables and benchmark calibration parameters.

Economic message: The figure and table together show the paper's central architecture: corporate risk management is a system of connected financial and real policies, not only a derivatives decision.

13.2 Figure 2: Case I–Liquidation and Figure 3: Case II–Optimal Refinancing

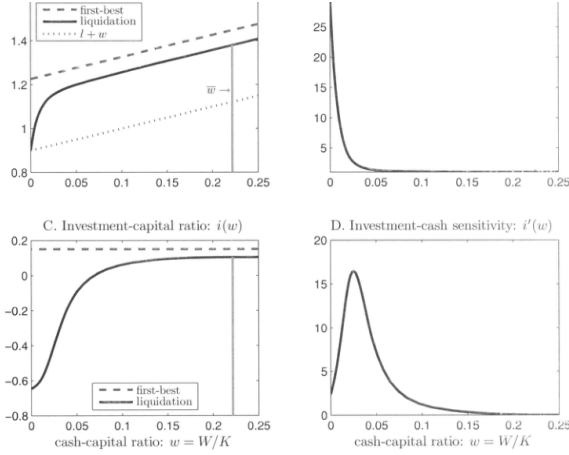


Figure 2. Case I—liquidation. This figure plots the solution for the case in which the firm has to liquidate when it runs out of cash ($w = 0$).

would permanently destroy the firm's future growth opportunities. Such high marginal value of cash highlights the importance of cash in periods of extreme financing frictions, which is what we have witnessed in the recent financial

(a) Figure 2: Case I—liquidation

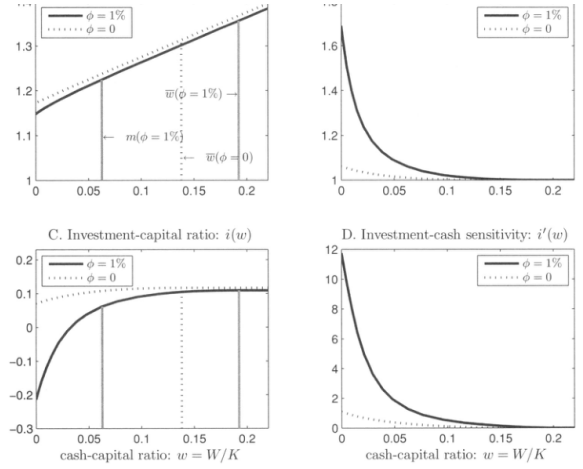


Figure 3. Case II—optimal refinancing. This figure plots the solution for the case of refinancing.

lies above $1 + \gamma$, reflecting the fact that the fixed cost component in raising

(b) Figure 3: Case II—optimal refinancing

Read:

- Figure 2 shows that liquidation risk produces an extremely high marginal value of cash near zero cash and severe asset sales.
- Figure 3 shows that refinancing softens the constraint but does not eliminate it when fixed issuance costs exist.
- In both cases, investment increases with the cash-capital ratio.

Economic message: The comparison isolates the role of external financing access. If no refinancing is possible, cash becomes survival capital. If refinancing is possible but costly, cash is still valuable because it helps avoid lumpy equity issuance.

13.3 Figure 4: Average q and Marginal q and Figure 5: Stationary Distributions in the Case of Refinancing

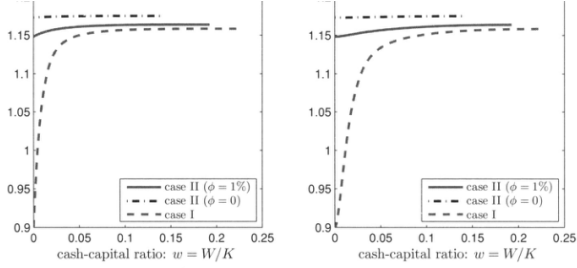
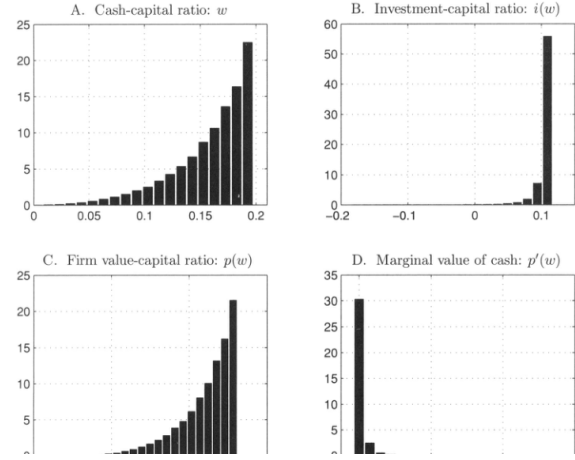


Figure 4. Average q and marginal q . This figure plots the average q and marginal q from the three special cases of the model. Case I is the liquidation case. Cases II and III are with external financing. The right end of each line corresponds to the respective payout boundary, beyond which both q_a and q_m are flat.

(a) Figure 4: Average q and marginal q



(b) Figure 5: Stationary distributions in the case of refinancing

Read:

- Figure 4 compares average and marginal q under liquidation and refinancing cases.
- Figure 5 shows that the stationary distribution puts much probability mass near high cash and high investment states.
- The firm endogenously avoids low-cash states most of the time.

Economic message: Observed firms may look only mildly constrained because they self-insure with cash. However, the tail states discipline the whole policy function.

13.4 Table II: Moments from the Stationary Distribution of the Refinancing Case and Figure 6: Optimal Hedging

Moments from the Stationary Distribution of the Refinancing Case

This table reports the population moments for the cash-capital ratio (w), the investment-capital ratio ($i(w)$), the marginal value of cash ($p'(w)$), average q ($q_a(w)$), and marginal q ($q_m(w)$) from the stationary distribution in the case with refinancing ($\phi = 1\%$).

	Cash-Capital Ratio w	Investment-Capital Ratio $i(w)$	Marginal Value of Cash $p'(w)$	Average q $q_a(w)$	Marginal q $q_m(w)$	Investment-Cash Sensitivity $i'(w)$
Mean	0.159	0.104	1.006	1.164	1.163	0.169
Median	0.169	0.108	1.001	1.164	1.164	0.064
Std.	0.034	0.013	0.018	0.001	0.001	0.376
Skewness	-1.289	-6.866	9.333	-8.353	-3.853	8.468
Kurtosis	4.364	76.026	146.824	106.580	22.949	122.694

The analysis above highlights that, by providing a more complete picture of

(a) Table II: Moments from the stationary distribution of the refinancing case

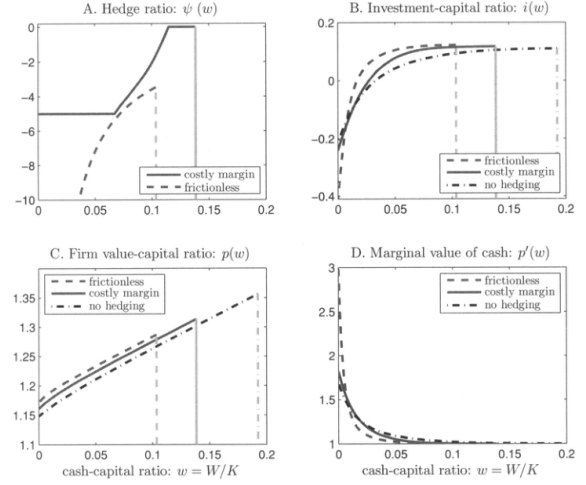


Figure 6. Optimal hedging. This figure plots the optimal hedging and investment policies, the firm value-capital ratio, and the marginal value of cash for Case II with hedging (with or without margin requirements). In Panel A, the hedge ratio for the frictionless case is cut off at -10 for

(b) Figure 6: Optimal hedging

Read:

- Table II reports that the marginal value of cash has mean close to one but very high skewness and kurtosis.
- Figure 6 shows hedging and investment policies with and without margin requirements.
- Hedging improves firm value, but margin requirements limit hedging when cash is scarce.

Economic message: Averages hide financing constraints. Tail risk and collateral constraints determine the value of cash and hedging even when ordinary states look close to first-best.

13.5 Figure 7: Credit Line and Figure 8: Investment and q with Credit Line

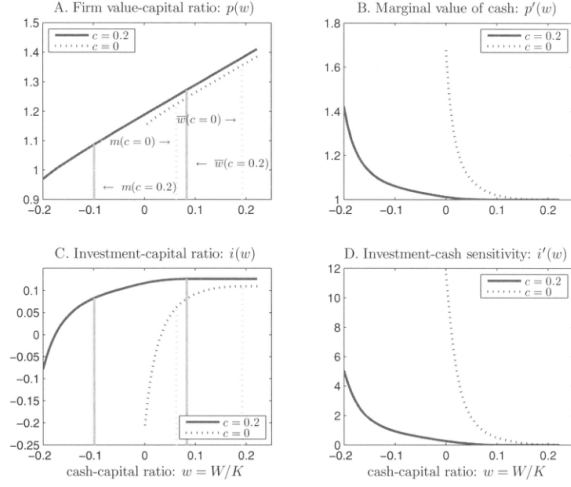


Figure 7. Credit line. This figure plots the model solution with credit lines and external equity. Each panel plots two scenarios: one without a credit line ($c = 0$) and the other with a credit line ($c = 20\%$). The spread on the credit line is $\alpha = 1.5\%$ over the risk-free rate r .

(a) Figure 7: Credit line

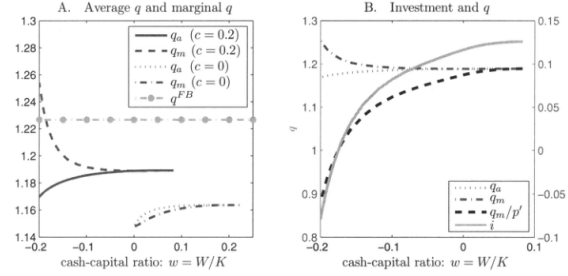


Figure 8. Investment and q with credit line. The left panel plots the average q (q_a) and marginal q (q_m) from the case with a credit line ($c = 0.2$) and without a credit line ($c = 0$) up to the respective payout boundaries. The right panel plots average q , marginal q , and the ratio of marginal q to the marginal value of liquidity (q_m/p') on the left axis, and the investment-capital ratio (i) on the right axis. These results are for the case with a credit line ($c = 0.2$).

(b) Figure 8: Investment and q with credit line

Read:

- Figure 7 shows that credit lines lower the marginal value of cash and reduce underinvestment.
- Figure 8 shows that marginal q and investment can move in opposite directions in the credit region.
- Average q tracks investment more robustly in the presence of credit-line financing.

Economic message: Credit lines change both financing policy and the empirical meaning of q . Investment is governed by liquidity-adjusted marginal q , not by marginal q alone.

14 Empirical and Research Implications

14.1 Implication 1: Investment-cash flow sensitivity is not a simple measure of constraints

Because firms endogenously manage cash to avoid constrained states, observed investment-cash flow sensitivities can be low even when financing constraints are important. The model suggests that tail behavior, cash distribution, issuance behavior, and payout boundaries matter for identifying constraints.

14.2 Implication 2: Cash holdings should be studied dynamically

A firm's cash ratio is not just a target level. It is part of an optimal band policy. Empirical cash regressions should distinguish the payout boundary, refinancing boundary, and the interior dynamics of cash accumulation.

14.3 Implication 3: Risk management should include operating decisions

The model predicts that firms manage risk using several tools at once: cash reserves, hedging, investment adjustment, asset sales, payout timing, and credit lines. Empirical work that studies derivatives without cash or cash without hedging misses an important complementarity.

14.4 Implication 4: Credit lines affect the interpretation of q

Credit lines make marginal q partly reflect debt-capacity effects. Therefore, the relationship between q and investment depends on whether the firm is financing at the cash margin or credit margin.

15 Critical Evaluation

15.1 Strengths

- The model unifies investment, cash, financing, payout, hedging, and credit-line decisions in one tractable framework.
- It provides clean analytical policy functions and clear quantitative predictions.
- The cash-capital ratio state variable gives an intuitive and operational measure of financial slack.
- The paper clarifies why financial constraints can be important even when firms usually appear unconstrained.

15.2 Limitations

- Investment opportunities are constant in the baseline model, so the paper abstracts from time-varying growth options.
- Financing costs are reduced-form rather than derived from explicit asymmetric information or agency frictions.
- The fixed equity issuance cost is proportional to firm size, which is analytically convenient but may not fit start-ups or very small firms.
- Long-term debt and debt maturity are not modeled in the main analysis.

15.3 Best way to remember the paper

The paper is a liquidity-adjusted q theory of investment. In a financially constrained firm, investment is not simply about how valuable capital is; it is about how valuable capital is relative to how valuable cash is.

16 Conclusion

Bolton, Chen, and Wang (2011) develop a dynamic theory in which the marginal value of liquidity coordinates corporate investment, cash, external financing, payout, hedging, and credit-line

decisions. The paper's most important insight is that financing constraints transform q theory: investment depends on marginal q divided by the marginal value of liquidity. The model also shows that cash management and financial hedging are complementary, and that credit lines can weaken the monotonic link between marginal q and investment. Overall, the paper provides a powerful framework for thinking about corporate policy in financially constrained firms, especially during crises when liquidity becomes extremely valuable.